PHYSICS

1) Find the projection of $\vec{A} = \hat{i} - \hat{j}$ along $\vec{B} = -\hat{i} + \hat{j}$:-

(1) $\frac{-1}{\sqrt{2}}$
(2) $-\sqrt{2}$
(3) $\sqrt{2}$
(4) None of these

2) The angle made by the vector $\vec{r} = \hat{i} + \hat{j}$ with x-axis is :-

(1) 30°
(2) 45°
(3) 60°
(4) 90°

3) If $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and $\left|\hat{a} + \hat{b}\right| = \sqrt{3}$, then the value of $(2\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$ is :-

(1) $\frac{41}{2}$
(2) $\frac{11}{2}$
(3) $-\frac{11}{2}$
(4) $-\frac{41}{2}$

4) If two vectors $(2\hat{i} + 3\hat{j} - \hat{k})$ and $(-4\hat{i} - 6\hat{j} + \lambda\hat{k})$ are parallel to each other then λ =

(1) 4
(2) 3
(3) 2
(4) 1

5) $\vec{A}$ and $\vec{B}$ are vectors expressed as $\vec{A} = 2\hat{i} + \hat{j}$ and $\vec{B} = \hat{i} - \hat{j}$. Unit vector perpendicular to $\vec{A}$ and $\vec{B}$ is :-

(1) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$
(2) $\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

(3) $\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

(4) $\hat{k}$

6) A particle moves in space according to equation $\vec{r} = (\sin t\,\hat{i} + \cos t\,\hat{j} + t\,\hat{k})m$.
Find time 't' when position vector and acceleration vector are perpendicular to each other :-

(1) 1

(2) $\frac{\pi}{4}$

(3) always

(4) never

7) Position of a particle moving along x–axis is given by x = $(8t - 4t^2)$ m. The distance travelled by the particle from t = 0s to t = 2s is :-

(1) 0

(2) 8 m

(3) 12 m

(4) 16 m

8) The displacement of a body is given to be proportional to the cube of time elapsed. The magnitude of the acceleration of the body is :-

(1) Increasing with time

(2) Decreasing with time

(3) Constant but not zero

(4) Zero

9) An ant is at a corner of a cubical room of side a. The ant can move with a constant speed u. The minimum time taken to reach the farthest corner of the cube is :

(1) $\frac{3a}{u}$

(2) $\frac{\sqrt{3}a}{u}$

(3) $\frac{\sqrt{5}a}{u}$

(4) $\frac{(\sqrt{2}+1)a}{2}$

10) A particle has an initial velocity of $3\hat{i} + 4\hat{j}$ and an acceleration of $0.4\hat{i} + 0.3\hat{j}$. It's speed after 10 sec is :-

(1) 7 units

(2) 8.5 units

(3) 10 units

(4) $7\sqrt{2}$ units

11) A body slides on an inclined plane. If height of inclined plane is 'h' and length of inclined plane is 'l' and angle of inclination is θ then time taken for travelling from upper point to lower point of incline :-

(1) $\sqrt{\frac{2h}{g}}$

(2) $\sqrt{\frac{2\ell}{g}}$

(3) $\frac{1}{\sin\theta}\sqrt{\frac{2h}{g}}$

(4) $\sin\theta\sqrt{\frac{2h}{g}}$

12) Two cars A and B at rest at same point initially. If A starts with uniform velocity of 40 m/sec and B starts in the same direction with constant acceleration of $4 m/s^2$, then B will catch A after how much time :-

(1) 10 sec
(2) 20 sec
(3) 30 sec
(4) 35 sec

13) A bullet moving with a velocity of 200 cm/s penetrates a wooden block and comes to rest after traversing 4 cm inside it. What initialvelocity is needed for travelling distance of 9 cm in same block:-

(1) 100 cm/s
(2) 150 cm/s
(3) 300 cm/s
(4) 250 cm/s

14) Two trains one of length 100 m and another of length 125 m, are moving in mutually opposite directions along parallel lines. They meet each other when each train has speed of 10 m/s. If their acceleration are $0.3 m/s^2$ and $0.2 m/s^2$ respectively, then the time they take to pass each other will be :-

(1) 5 s
(2) 10 s
(3) 15 s
(4) 20 s

15) Velocity vector of a particle is given as $\vec{V} = 4t\hat{i} + 3\hat{j}$. At t = 0 position of the particle is given as

$\vec{r}_0 = 2\hat{j}$. Find the position vector of the particle at t = 2 sec. :-

(1) $8\hat{i} + 6\hat{j}$
(2) $8\hat{i} + 8\hat{j}$
(3) $8\hat{i} + 4\hat{j}$
(4) None

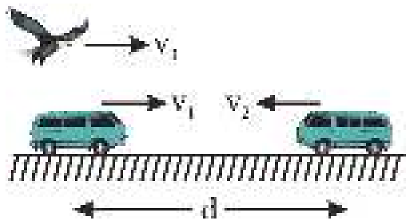
16) A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity u at an angle $\left(\frac{\pi}{2} - \theta\right)$ with the horizontal, it reaches maximum height H_2.The relation between the horizontal range R of the projectile, H_1 and H_2 is :–

(1) $R = 4\sqrt{H_1 H_2}$
(2) $R = 4(H_1 - H_2)$
(3) $R = 4(H_1 + H_2)$
(4) $R = \frac{H_1^2}{H_2^2}$

17) A man is crossing a river flowing with velocity of 5m/s. He reaches a point directly across at a distance of 60m in 5 sec. His velocity in still water should be :-

(1) 12 m/s
(2) 13 m/s
(3) 5 m/s
(4) 10 m/s

18) A bird flies to and fro between two cars which move with velocities v_1 = 20 m/s and v_2 = 30 m/s. If the speed of the bird is v_3 = 10 m/s and the initial separation between cars d = 2 km, find the total distance covered by the bird till the cars meet :-



(1) 2000 m
(2) 1000 m
(3) 400 m
(4) 200 m

19) A 210 m long train is moving due North at a speed of 25 m/s. A small bird is flying due South a little above the train with speed 5 m/s. The time taken by the bird to cross the train is :-

(1) 6s
(2) 7s
(3) 9s

(4) 10s

20) A projectile is thrown with an initial velocity of $\vec{v} = a\,\hat{i} + b\,\hat{j}$. if range of the projectile is double the maximum height attained by it then :

(1) a = 2 b
(2) b = a
(3) b = 2a
(4) b = 4a

21) A boat is sailing with a velocity $(3\hat{i} + 4\hat{j})$ with respect to ground and water in river is flowing with a velocity $(-3\hat{i} - 4\hat{j})$. Relative velocity of the boat with respect to water is :

(1) $8\hat{j}$
(2) $5\sqrt{2}$
(3) $6\hat{i} + 8\hat{j}$
(4) $-6\hat{i} - 8\hat{j}$

22) A river is flowing from W to E with a speed of 5 m/min. A man can swim in still water with a velocity 10 m/min. In which direction should the man swim so as to take the shortest possible path to go to the south :-

(1) 30° with downstream
(2) 60° with downstream
(3) 120° with downstream
(4) South

23) The phase difference between the instantaneous velocity and acceleration of a particle executing simple harmonic motion is :-

(1) π
(2) 0.707 π
(3) zero
(4) 0.5 π

24) Displacement-time equation of a particle executing SHM is $x = A\sin\left(\omega t + \frac{\pi}{6}\right)$.Time taken by the particle to go directly from

$x = -\frac{A}{2}$ to $x = +\frac{A}{2}$ is :

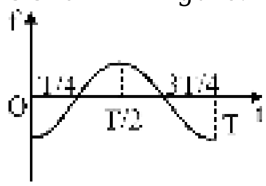
(1) $\frac{\pi}{3\omega}$
(2) $\frac{\pi}{2\omega}$
(3) $\frac{2\pi}{\omega}$

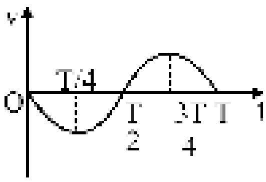
(4) $\frac{\pi}{\omega}$

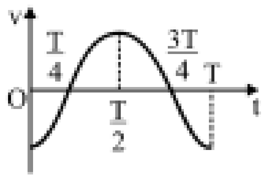
25) A particle excutes SHM on a straight line path. The amplitude of oscillation is 2 cm. When the displacement of the particle from the mean position is 1 cm, the numerical value of magnitude of acceleration, is equal to the numerical value of magnitude of velocity. The frequency of SHM (in second^{-1}) is :-

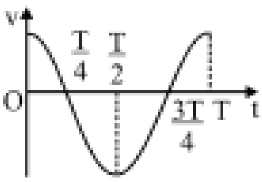
(1) $2\pi\sqrt{3}$
(2) $\frac{2\pi}{\sqrt{3}}$
(3) $\frac{\sqrt{3}}{2\pi}$
(4) $\frac{1}{2\pi\sqrt{3}}$

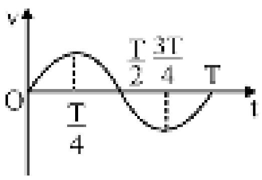
26) A body is performing simple harmonic motion with amplitude a and time period T. Variation of its acceleration (f) with time (t) is shown in figure. If at time t, velocity of the body is v, which of the following v-t graphs is correct ?



(1)


(2)


(3)


(4)


27) On suspending a mass M from a spring the period of oscillations is T. It is cut into n equal segments on suspending the mass M from one of the segments. On the time period become $\frac{T}{\sqrt{3}}$, then the value of n will be :-

(1) 9

(2) 2

(3) 3

(4) None

28) A particle executes SHM of period 8 seconds. After what time of its passing through mean position, will the energy be half kinetic and half potential ?

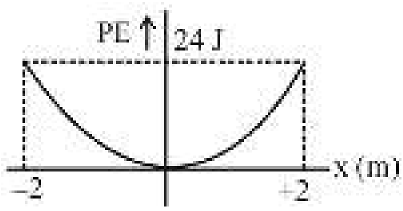
(1) 2 sec

(2) 1 sec

(3) 2.5 sec

(4) 4 sec

29) A particle is executing S.H.M. Its potential energy v/s displacement graph is given below.



The value of restoring force constant is :-

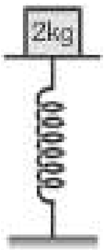
(1) 12 N/m

(2) 24 N/m

(3) 6 N /m

(4) 48 N/m

30) A mass of 2.0 kg is put on a flat pan attached to a vertical spring fixed on the ground as shown in the figure. The mass of the spring and the pan is negligible. When pressed slightly and released, the mass executes a simple harmonic motion. The spring constant is 200 N/m. What should be the minimum amplitude of the motion so that the mass gets detached from the pan : (Take, g = 10 m/s^2)



(1) 10.0 cm

(2) any value less than 12.0 cm

(3) 4.0 cm

(4) 8.0 cm

31) A particle executes simple harmonic motion with a time period of 16 s. At time t = 2s, the particle crosses the mean position while at t = 4s, its velocity is 4 ms^{-1}. The amplitude of motion (in metre) is :

(1) $\frac{\sqrt{2}}{\pi}$

(2) $\frac{16\sqrt{2}}{\pi}$

(3) $\frac{24\sqrt{2}}{\pi}$

(4) $\frac{32\sqrt{2}}{\pi}$

32) A particle, moving on x-axis, has potential energy $U = 2 - 20x + 5x^2$ joule. The particle is released at $x = -3$. The maximum value of 'x' will be [x is in meter and U is in joule] :-

(1) 5 m
(2) 3 m
(3) 7 m
(4) 8 m

33) A transverse periodic wave on a string with a linear mass density of 0.200 kg/m is described by the equation $y = 0.05 \sin(420t - 21.0x)$ where x and y are in metres and t is in seconds. The tension in the string is equal to :

(1) 32 N
(2) 42 N
(3) 66 N
(4) 80 N

34) A wave in a string has an amplitude of 2cm. The wave travels in the + ve direction of x axis with a speed of 128 m/sec and it is noted that 5 complete waves fit in 4 m length of the string. The equation describing the wave is:

(1) y = (0.02) m sin (7.85x – 1005t)
(2) y = (0.02) m sin (7.85x + 1005t)
(3) y = (0.02) m sin (15.7x – 2010t)
(4) y = (0.02)m sin (15.7x + 2010t)

35) A heavy rope is suspended from a rigid support. A wave pulse is set up at lower end, then :-

(1) the pulse will travel with uniform speed
(2) the pulse will travel with increasing speed
(3) the pulse will travel with decreasing speed
(4) the pulse can not travel through rope

36) Two waves of same frequency and of intensity I_0 and $9I_0$ produces interference. If at a certain point the resultant intensity is $7I_0$, then the minimum phase difference between the two sound waves at that point will be

(1) 90°
(2) 100°
(3) 120°

(4) 110°

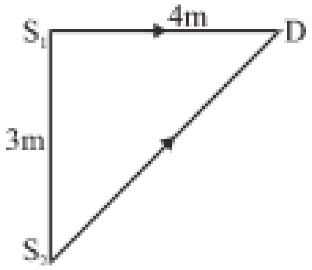
37) At which temperature the speed of sound will be three times of its speed at 0°C ?

(1) 1100°C
(2) 1284°C
(3) 1500°C
(4) 2184°C

38) The bob of a simple pendulum is a spherical hollow ball filled with water. A plugged hole near the bottom of the oscillating bob gets suddenly unplugged. During observation, till water is coming out, the time period of oscillation would:-

(1) remain unchanged
(2) increase towards a saturation value
(3) first increase and then decrease to the original value
(4) first decrease and then increase to the original value

39) In the figure, the intensity of waves arriving simultaneously at point D from two coherent sources S_1 & S_2 is I_0 each. The wavelength of the wave is λ = 4m. Resultant intensity at D will be :



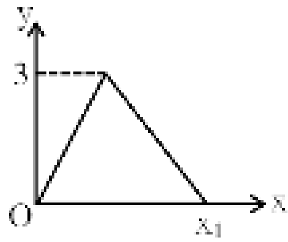
(1) 4 I_0
(2) I_0
(3) 2 I_0
(4) Zero

40)

Find minimum value of

$y = 5x^2 - 2x + 1$ is

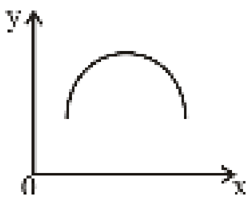
(1) $\frac{4}{5}$
(2) $\frac{3}{5}$
(3) $\frac{1}{5}$
(4) $\frac{2}{5}$

41) Find the value of x_1 so that $\int_0^{x_1} y\,dx = 15$ 

(1) 5
(2) 10
(3) 15
(4) 20

42) The equation of a curve is given as $y = x^2 + 2 - 3x$. The curve intersects the x-axis at

(1) (1, 0)
(2) (2, 0)
(3) Both (1) and (2)
(4) No where

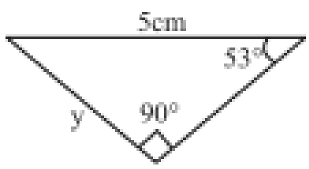
43) 

For the given curve

(1) slope is continuously decreasing
(2) slope is continuously increasing
(3) slope is constant
(4) slope is always positive

44) Value of $(1.002)^3$ = ...

(1) 1.008
(2) 1.004
(3) 1.006
(4) 1.005

45) Find out perimeter of triangle 

(1) 14 cm
(2) 12 cm
(3) 10 cm

(4) 8 cm

CHEMISTRY

1) Which of the following molecules has the maximum number of A–X bonds of identical length, where 'A' is the central atom and 'X' is the surrounding atom :-

(1) SF_6

(2) IF_7

(3) PCl_5

(4) ClO_4^-

2) **Assertion (A) :-** F_2 has two unpaired electrons in π* ABMO. **Reason (R) :-** F_2 is coloured due to HOMO-LUMO transition.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

3) ***Assertion (A)*** :- During the formation of O_2^+ from O_2, stability increases.
Reason (R) :- e^- is removed from Antibonding Molecular orbitals.

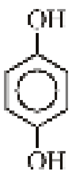
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

4) Which of the following molecule has some dipole moment :-

(1) 

(2) XeO_2F_2

(3) PF_3Cl_2

(4) All

5) Which of the following pair of species have identical shapes ?

(1) NO_2^+ and NO_2^-

(2) PCl_5 and BrF_5

(3) XeF_4 and ICl_4^-

(4) $TeCl_4$ and XeO_4

6)

How many of the following compounds are paramagnetic ?
KO_2, H_2O_2, BaO_2, O_3, O_2, $O_2[PtF_6]$

(1) 3
(2) 4
(3) 5
(4) 6

7) Number of compounds having X–O–X linkage
N_2O_5, Cl_2O_7, S_3O_9, $Na_2B_4O_7.10H_2O$, P_4O_{10}, P_4O_6, N_2O_4, Cl_2O_6

(1) 7
(2) 6
(3) 5
(4) 8

8) Which of the following d-orbital is used in the hybridisation of P in anionic part of $PCl_5(s)$.

(1) d_{yz}
(2) $d_{x^2-y^2}$
(3) d_{z^2}
(4) Option (2) and option (3) both

9) The number of F–I–F angles less than 90° and equal to 90° in IF_7 are respectively :

(1) 5, 5
(2) 10, 5
(3) 5, 10
(4) 10, 15

10) In which of the following statement is/are correct -

(1) C-X bond length order ⇒ $CH_3Cl < CH_3F$
(2) In PCl_3F_2 , $P–F_a$ bond length is greater than $P–Cl_e$ bond length.
(3) $X-\hat{C}-X$ bond angle ⇒ $OCF_2 < OCCl_2$
(4) None of these

11)

Consider the following four xenon compounds: XeF_4, XeF_2, XeO_3, XeO_2F_2. The pair of xenon compounds expected to have non-zero dipole moment is

(1) XeF_4 and XeO_3
(2) XeO_3 and XeF_2

(3) XeO_3 and XeO_2F_2

(4) XeO_2F_2 and XeF_4

12) Total number of right angles $\angle ClPCl$ present in PCl_5, PCl_4^+, PCl_6^- are

(1) 0, 1, 4

(2) 6, 0, 4

(3) 2, 4, 0

(4) 6, 0, 12

13) In which of the following orbital having lone pair of underlined atom have maximum s-character ?

(1) PF_3

(2) PCl_3

(3) PH_3

(4) NH_3

14) Choose incorrect order.

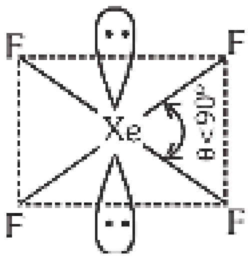
(1) Bond angle order : $NH_3 > PH_3 > AsH_3 > SbH_3$

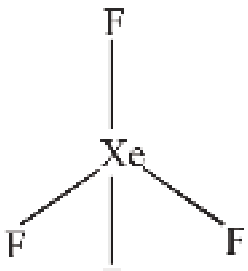
(2) Bond length order : $F_2 < Cl_2 < Br_2 < I_2$

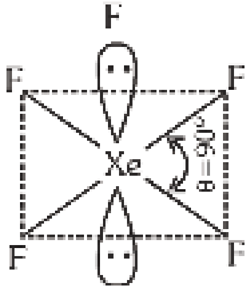
(3) Bond strength order : $F_2 > Cl_2 > Br_2 > I_2$

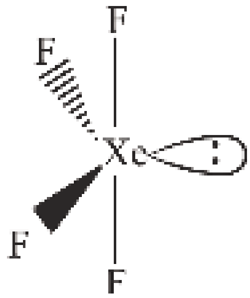
(4) Order of dipole moment : $NH_3 > NF_3 > BF_3$

15) Select best theory representation of XeF_4 according to VSEPR :-

(1) 

(2) 

(3) 

(4) 

16) Which of the following odd electron species planar and polar.

(1) $\dot{C}H_3$

(2) $\dot{C}F_3$

(3) NO_2

(4) ClO_3

17) Select the incorrect statement(s) :

(1) PF_5 exist where as PH_5 does not exist.

(2) In PCl_2F_3, P – Cl equitorial bond length is more than P–F axial bond.

(3) Hybridisation of phosphorous in PCl_5 is sp^3.

(4) ICl_7 does not exist due to steric hinderance.

18)

Match column-I with column–II and select the correct answer :-

Column-I (Molecules)		**Column-II (Property)**	
(P)	IF_7	(1)	Planar and Polar
(Q)	SO_2	(2)	Non planar and polar
(R)	SF_4	(3)	Planar and Nonpolar
(S)	CS_2	(4)	Non polar and non planar

	(P)	(Q)	(R)	(S)
(1)	4	1	2	3
(2)	4	2	1	3
(3)	3	1	2	4
(4)	3	1	4	2

(1) 1

(2) 2

(3) 3

(4) 4

19) Shape of which of the following molecule can be explained without considering hybridisation?

(1) AsH_3
(2) H_2O
(3) NH_3
(4) All of these

20) Which of the following order is **CORRECT** for their indicated properties ?

(1) H-C-H bond angle order : $CH_3 > CH_4$
(2) Order of %s character in Xe-O bond : $XeO_3 > XeO_4$
(3) % s-character in the orbital containing lone pair : $PH_3 < PF_3$
(4) % s-character in E – H bonds (E = central atom) : $NH_3 < AsH_3$

21) In which of the following species pπ – pπ bond is absent ?

(1) SO_3
(2) NO_3^-
(3) SO_4^{-2}
(4) CO_3^{-2}

22) If pure unhybridised orbitals are used in forming CH_4 then which statement is incorrect :-

(1) Three bond angle will be at 90°.
(2) Three bond lengths are identical and fourth is smaller than other three.
(3) Geometry will be tetrahedral.
(4) Two types of atomic orbitals will be used by carbon.

23) In PO_4^{-3} ion the average formal charge on the oxygen atom is :-

(1) +1
(2) –1
(3) $-\frac{3}{4}$
(4) $+\frac{3}{4}$

24) **Assertion :** The dipole moment of O_2F_2 is not equal to zero.
Reason : All atom are lying is same plane.

(1) Both **Assertion** and **Reason** are true and **Reason** is correct explanation for **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is not correct explanation for **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

25) Which of the following order for bond length is correct ?

(1) d_{N-N} in $N_2H_4 < d_{N-N}$ in N_2F_4
(2) d_{C-C} in $C_2H_4 < d_{C-C}$ in C_2F_6
(3) d_{O-O} in $O_2H_2 < d_{O-O}$ in O_2F_2
(4) None of these

26) For the reaction xA + yB → zC
$-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = 1.5\frac{d[C]}{dt}$
then x, y and z are respectively :-

(1) 1, 1, 1
(2) 3, 2, 3
(3) 3, 3, 2
(4) 2, 2, 3

27) Rate constant for the reaction is 1.5×10^7 sec^{-1} at 50°C and 4.5×10^7 sec^{-1} at 100°C.What is the value of activation energy ?

(1) 2.2×10^2 J/mole
(2) 2.2×10^5 J/mole
(3) 2.2×10^3 J/mole
(4) 2.2×10^4 J/mole

28) For a zero order reaction. Which of the following statement is false :

(1) the rate of reaction is independent of the temperature
(2) the rate is independent of the concentration of the reactants.
(3) the half life depends upon the concentration of the reactants.
(4) the rate constant has the unit mole L^{-1} s^{-1}.

29) The hypothetical reaction $P_2 + Q_2 \rightarrow 2PQ$ follows the mechanism as given below :
$P_2 \rightleftharpoons 2P$ (fast)
$P + Q_2 \rightarrow PQ + Q$ (slow)
$P + Q \rightarrow PQ$ (fast)
The order of the overall reaction is :-

(1) 2
(2) 1
(3) 1.5
(4) 0

30) $A_2 + B_2 \rightarrow 2AB$; R.O.R. = $k[A_2]^a [B_2]^b$

Initial $[A_2]$	**Initial $[B_2]$**	**R.O.R. (r) Ms^{-1}**
0.2	0.2	0.04
0.1	0.4	0.04

0.2	0.4	0.08

Order of reaction with respect to A_2 and B_2 are respectively :

(1) a = 1, b = 1

(2) a = 2, b = 0

(3) a = 2, b = 1

(4) a = 2, b = 2

31) The half life period and initial concentration of reactant for a reaction are as follows. What is order of reaction ?

Initial concentration (M))	0.5	1.0	2.0
$t_{1/2}$ (hr)	3.64	1.82	0.91

(1) zero

(2) 1

(3) 2

(4) 3

32) In the given first order reaction
$NH_4NO_{2(aq.)} \rightarrow N_{2(g)} + 2H_2O_{(l)}$ the volume of N_2 after 20 min and after a long time is 40 mL and 70 mL respectively. The value of rate constant is:-

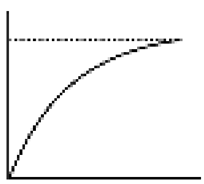
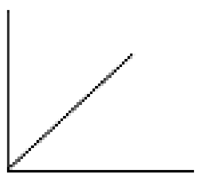
(1) (1/20) ln (7/4) min^{-1}

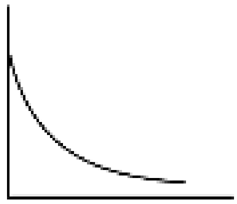
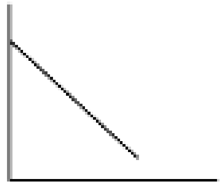
(2) (2.303/1200) log (7/3) sec^{-1}

(3) (1/20) log (7/3) min^{-1}

(4) (2.303/20) log (11/7) min^{-1}

33) Match the entry of Column-I with entries of Column-II.

Column-I (First order reaction A→B)		**Column-II**	
(A)	[B] vs time	(P)	
(B)	Rate vs [A]	(Q)	

(C)	[A] vs time	(R)	
(D)	ℓn[A] vs time	(S)	

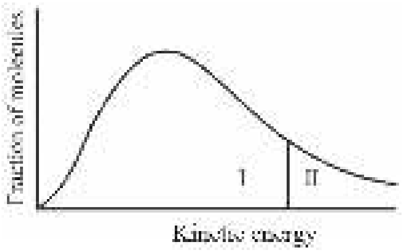
(1) A→Q; B→S; C→P; D→R

(2) A→P; B→Q; C→R; D→S

(3) A→P; B→R; C→Q; D→S

(4) A→P; B→S; C→Q; D→R

34) Consider the following graph of the kinetic energy distribution among molecules at temperature



T. If the temperature is increased, how would the resulting graph differ from the above one :

(1) Both area I and II would increase.

(2) Both area I and II would decrease.

(3) Area I would increase and area II would decrease

(4) Area I would decrease and area II would increase

35)

In the following reaction A → B + C, rate constant is 0.001 Ms^{-1}. If we start with 1M concentration of A, then conc. of A and B after 10 minutes are respectively :-

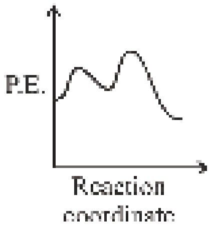
(1) 0.5 M, 0.5 M

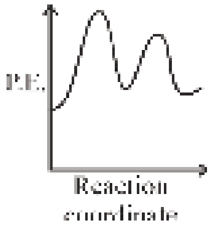
(2) 0.6 M, 0.4 M

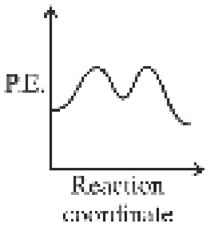
(3) 0.4 M, 0.6 M

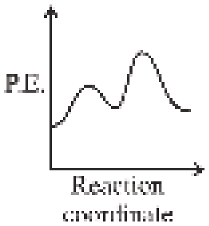
(4) None of these

36) Select the correct diagram for an endothermic reaction that proceeds through two steps, with the second step is rate determining :-

(1) 

(2) 

(3) 

(4) 

37)

For gaseous reactions, following data is given

$A \rightarrow B \; K_1 = 10^{16} e^{-2000/T}$,

$C \rightarrow D \; K_2 = 10^{14} e^{-500/T}$

The temperature at which $K_1 = K_2$:-

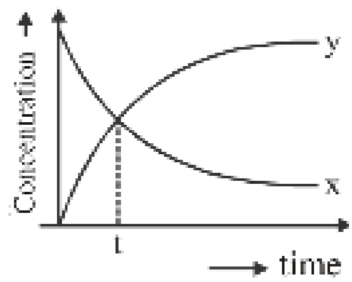
(1) 325.7 K

(2) 200 K

(3) 651.4 K

(4) 1000 K

38) For the reaction $x \rightarrow 2y$, following curve is obtained, then time 't' is :



(1) $t_{2/3}$

(2) $t_{1/4}$

(3) $t_{1/2}$

(4) $t_{1/3}$

39) Match the rate law given in column I with the dimensions of rate constants given in column II and mark the appropriate choice. :-

Column-I		Column-II	
(A)	Rate = $k[NH_3]^0$	(i)	mol L^{-1} s^{-1}
(B)	Rate = $k[H_2O_2][I^-]$	(ii)	L $mol^{-1}s^{-1}$
(C)	Rate = $k[CH_3CHO]^{3/2}$	(iii)	s^{-1}
(D)	Rate = $k[C_2H_5Cl]$	(iv)	$L^{1/2}$ $mol^{-1/2}$ s^{-1}

(1) (A) → (iv), (B) → (iii), (C) → (ii), (D) → (i)

(2) (A) → (i), (B) → (ii), (C) → (iii), (D) → (iv)

(3) (A) → (ii), (B) → (i), (C) → (iv), (D) → (iii)

(4) (A) → (i), (B) → (ii), (C) → (iv), (D) → (iii)

40) Match the column :-

	Column-I		Column-II
(A)	Radioactive decay	(P)	Pseudo first order
(B)	Decomposition of HI on gold surface at high pressure	(Q)	First order
(C)	Saponification	(R)	Zero order
(D)	Inversion of cane sugar in excess of water	(S)	Second order

(1) (A) → (Q), (B) → (R), (C) → (S), (D) → (P)

(2) (A) → (R), (B) → (Q), (C) → (P), (D) → (S)

(3) (A) → (P), (B) → (Q), (C) → (R), (D) → (S)

(4) (A) → (S), (B) → (P), (C) → (Q), (D) → (R)

41) Given below are two statements :
Statements I :- To express the rate at a particular moment of time we determine the instantaneous rate.
Statements II :- Instantaneous rate is obtained when we consider the average rate at the smallest time interval dt (when Δt approaches zero).

(1) Both statement I and statement II are false

(2) Both statement I and statement II are true

(3) Statement I is true but statement II is false

(4) Statement I is false but statement II is true

42) 99% of a first order reaction was completed in 32 minutes when 99.9% of the reaction will complete :-

(1) 50 min
(2) 46 min
(3) 48 min
(4) 49 min

43) Rate of formation of SO_3 in the following reaction $2SO_2 + O_2 \rightarrow 2SO_3$ is 100 gm/min. Rate of disappearance of O_2 is:

(1) 50 gm/min
(2) 200 gm/min
(3) 40 gm/min
(4) 20 gm/min

44) **Assertion :** An elementary reaction cannot have fractional order.
Reason : Stoichiometric coefficients in an elementary reaction can be fractional.

(1) Assertion and Reason are true and Reason is correct explanation of Assertion.
(2) Assertion and Reason are true and Reason is not correct explanation of Assertion.
(3) Assertion is true and Reason is wrong.
(4) Assertion is wrong and Reason is true.

45) Half lives of a first order and a zero order reactions are same. Then the ratio of the initial rates of first order reaction to that of the zero order reaction is :
(Assume initial concentration of reactants are 1 Molar)

(1) $\frac{1}{0.693}$
(2) 2×0.693
(3) 0.693
(4) $\frac{2}{0.693}$

BIOLOGY

1) Young one of the cockroach is called

(1) Naid
(2) Grub
(3) Nymph
(4) Maggot

2) In cockroach a sac like structure present in alimentary canal and used for storing of food called :-

(1) Pharynx
(2) Gizzard
(3) Crop

(4) Stomach

3) Which structure represent the brain of cockroach?

(1) Supra–Pharyngeal ganglion
(2) Sub–Pharyngeal ganglion
(3) Supra–Oesophageal ganglion
(4) Sub–Oesophageal ganglion

4) The head of the cockroach is formed by the fusion of how many segments–

(1) Five
(2) Six
(3) Three
(4) Ten

5) Incorrect about cockroach–

(1) The adults of the common species of cockroach, Periplaneta americana are about 34-53 cm long with wings that extend beyond the tip of the abdomen in males.
(2) The body of the cockroach is segmented and divisible into three distinct regions – head, thorax and abdomen.
(3) The entire body is covered by a hard chitinous exoskeleton (brown in colour).
(4) Sclerites are joined to each other by a thin and flexible articular membrane (arthrodial membrane).

6) Main difference between male and female cockroach is of–

(1) Antennae
(2) Mandible
(3) Podomeres
(4) Anal style

7) In cockroach, the ootheca is formed by the secretion of :-

(1) Phallic gland
(2) Uricose gland
(3) Collaterial gland
(4) Mushroom gland

8) Choose the statement which is not correct for cockroach?

(1) Blood vascular system of cockroach is a closed type.
(2) The abdomen in both males and females consists of 10 segments.
(3) On an average female produce 9-10 oothecae.
(4) Malpighian tubules help in excretion.

9) Spiracles found in cockroach are :-

(1) 2 pairs in thorax and 8 pairs in abdomen
(2) 2 pairs in thorax and 10 pairs in abdomen
(3) 8 pairs in thorax and 2 pairs in abdomen
(4) 2 pairs in thorax and 4 pairs in abdomen

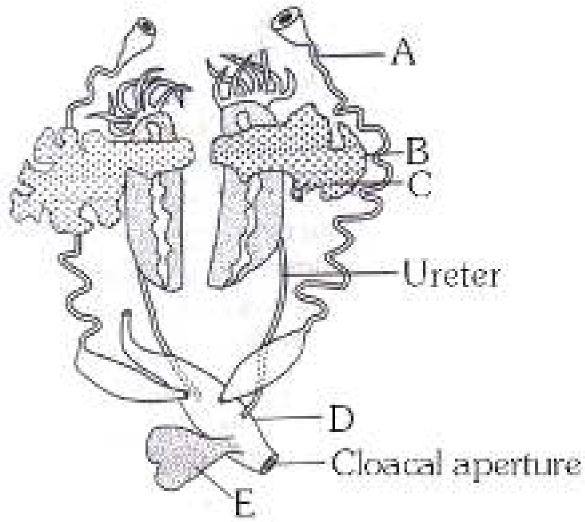
10) **Assertion :** Frogs exhibit sexual dimorphism.
Reason : Male frog can be distinguished by the presence of sound producing vocal sacs which is absent in female frogs.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

11) Given below is a list of some structure : Diaphragm, Tail, Anus, Cloaca, Nictitating membrane and the cortex and medulla in kidney. How many of the above structure are not found in *Rana tigrina* ?

(1) 3
(2) 5
(3) 6
(4) 4

12) Observe the following figure of female reproductive system of frog and identify A to E.



(1) A–Urinary duct, B-Ova, C–Ovary, D–Cloaca, E–Urethra
(2) A–Oviduct, B-Ovary, C–Ova, D–Cloaca, E–Urinary bladder
(3) A–Oviduct, B-Ovary, C–Ova, D–Rectum, E–Adrenal gland
(4) A–Urinogenital duct, B-Ovary, C–Ovum, D–Coelom, E–Urethra

13) Which Statement is true (about frogs) :

(1) Kidney is compact, light red and bean shape structures situated a little posteriorly in the body cavity on both sides of vertebral column
(2) Food is captured by the trilobed tongue. Digestion of food takes place by the action of HCl and gastric juices secreted from the walls of the stomach
(3) Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the second digit of the fore limbs which are absent in female frogs
(4) They have the ability of change the colour to hide them from their enemies. This protective coloration is called mimicry.

14) Frog has different types of sense organs.
(I) Sensory papillae
(II) Nasal epithelium
(III) Taste buds
(IV) Eyes
(V) Tympanum with internal ears
Which of these are well organzied structures?

(1) I and III
(2) III and IV
(3) IV and V
(4) I, II, III and IV

15) Which of the following respiration take place during hibernation of frog :

(1) Pulmonary respiration
(2) Cutaneous respiration
(3) Buccal cavity respiration
(4) Gill respiration

16) In frog, a triangular structure called .....A..... Joins the right atrium the ventricle opens into a sac like .....B.... on the ventral side of heart. Here A and B are respectively.

(1) A → Sinus Venosus; B → Conus arteriosus
(2) A → Conus arteriosus; B → Sinus Venosus
(3) A → Truncus arteriosus; B → Sinus Venosus
(4) A → Sinus Venosus; B → Systemic aorta

17) An epithelium tissue which has thin flat cells, appears like closely packed tiles is found to be present at ?

(1) Inner lining of stomach
(2) Inner lining of fallopian tube
(3) Inner lining of urinary bladder
(4) Bowman's Capsule

18) Stratified squamous Epithelium :-

(1) Outer most layer squamous & Inner most is cuboidal
(2) Outer most layer cuboidal & Inner most is squamous
(3) Outer most layer columnar & Inner most is cuboidal
(4) Outer most layer cuboidal & Inner most is columar

19) Which of the following are **correct** for epithelial tissue?
(A) It has free surface which may face towards the body fluid.
(B) It provides covering for some part of body
(C) Intercellular matrix is abundant
(D) It lies on basement membrane.

(1) (A), (B), (C), (D)
(2) Only (C)
(3) (A), (C), (D)
(4) (A), (B), (D)

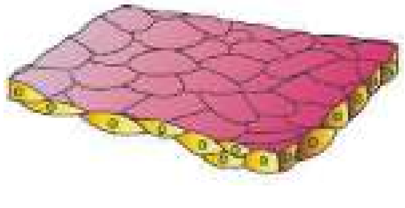
20) Tendons are :-
(A) Dense regular connective tissue with elastic fibres.
(B) Dense regular connective tissue with collagen fibres.
(C) Dense irregular connective tissue with collagen fibres.
(D) Dense irregular connective tissue with elastic fibres.

(1) Only (B)
(2) (B) and (C)
(3) (A) and (D)
(4) (A) and (C)

21) Which of the following statement is correct.

(1) matrix of cartilage semi solid, pliable and can resist compression
(2) Bone is mineralised, solid connective tissue having matrix rich in calcium salt and collagen fibers
(3) matrix of cartilage is secreted by osteocytes
(4) Tendon connects bone to bone

22) Identify the type of epithelial tissue shown below as well as the related right place of its occurrence in our body along with its correct function and select the correct option for the two together ?



	Type of epithelial Tissue	Occurance & its function

(1)	Simple squamous Epithelium	- Found in wall of blood vessels and air sacs of lungs. - Function-forming a diffusion boundary
(2)	Simple cuboidal epithelium	- Found in wall of ducts of gland - Function-diffusion and excretion
(3)	Simple columnar epithelium	- Found in wall of nephrones and wall of ducts of glands - Function-secretion and Absorption
(4)	Simple squamous epithelium	- Found in wall of stomach and intestine - Function-absorption and secretion

(1) 1
(2) 2
(3) 3
(4) 4

23) It serves as a support framework for epithelium. It contains fibroblast, macrophages and mast cells. It is known as.

(1) Dense regular tissue
(2) Adipose tissue
(3) Dense irregular tissue
(4) Areolar tissue

24)

Respiratory organ arthropoda is :-

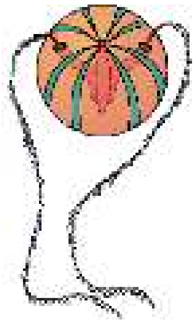
(1) Gills
(2) Book gills
(3) Book lungs
(4) 1,2 & 3

25) Which annelid is dioecious.

(1) Earthworm
(2) Leech
(3) Nereis
(4) Ascaris

26) An animal phylum having radially symmetrical adults but bilateral symmetrical larvae is :

(1) Porifera
(2) Coelenterata
(3) Echinodermata
(4) Annelida



27) Which of the following features is not correct for the above creature :

(1) Comb plates (External rows)
(2) Cnidoblast
(3) Sexes are not separate
(4) Digestion is both extracellular and intracellular

28) The body is externally and internally divided into segments with a serial repetition of at least some organs in

(1) Hydra
(2) Fasciola
(3) Earthworm
(4) Pila

29) Which of the following is limbless amphibian ?

(1) Hyla
(2) Bufo
(3) Salamandra
(4) Ichthyophis

30) Vertebrata is a subphylum and Pisces is a

(1) Class
(2) Super class
(3) Division
(4) Sub class

31) A zoologist came across a fish that has to swim constantly in water to avoid sinking. Identity that particular fish.

(1) Fighting fish

(2) Dog fish
(3) Angel fish
(4) Catla

32) Polyp phase is absent in........

(1) Hydra
(2) Aurelia
(3) Physalia
(4) Obelia

33) When any longitudinal plane passing through the central axis of the body divides the organism into two identical halves, it is called–

(1) Radial symmetry
(2) Bilateral symmetry
(3) Biradial symmetry
(4) Asymmetry

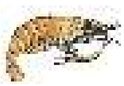
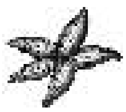
34) The long bones are hollow and connected by air passage. They are the characteristics of

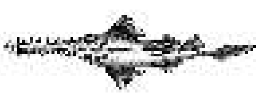
(1) Aves
(2) Mammalia
(3) Reptilia
(4) Land vertebrates

35) Some animals are given in the list below :
Asterias, Pila, Echinus, Antedon, Pinctada, Cucumaria, Octopus, Loligo, Ophiura, Dentalium
How many animals among these are related to the second largest phylum of animals?

(1) Two
(2) Three
(3) Four
(4) Five

36) Examine the given table with animal taxon is correctly match their characters with suitable examples :

(A)  (B) 

(C)  (D) 

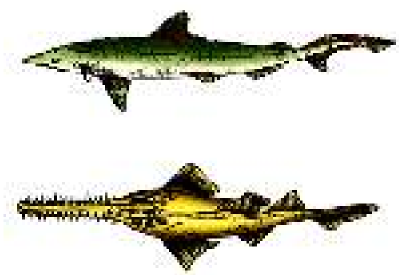
	A	B	C	D

(1)	Arthropoda rasping organ radula is present Ex. Prawn	Echinodermata spiny skin Ex. *Asterias*	Urochordata Marine animal Ex. *Ascidia*	Osteichthyes Bony endoskeleton Ex. *Scoliodon*
(2)	Arthropoda Chitinous exoskeleton Ex. Prawn	Echinodermata Sexual reproduction Ex. *Asterias*	Urochordata notochord is present in larval tail Ex. *Ascidia*	Chondrichthyes Cartilagenous endoskeleton Ex. - *Pristis*
(3)	Annelida Segmented body Ex. Scorpion	Mollusca Calcareous shell Ex. *Ophiura*	Cephalochordata Marine animal Ex. *Salpa*	Amphibia Oviparous Ex. *Salamendra*
(4)	Aschelminthes Common name roundworm Ex. *Prawn*	Arthropoda Malpighean tubules present Ex. Butter fly	Hemichordata Proboscis gland present Ex. *Saccoglossus*	Osteichthyes Operculum present Ex. *Labeo*

(1) 1
(2) 2
(3) 3
(4) 4

37) **Assertion :-** Hippocampus is an example of subphylum vertebrata.
Reason :- The members of subphylum vertebrata possess notochord during the embryonic period.

(1) Assertion is correct but reason is incorrect
(2) Both assertion and reason are incorrect
(3) Both assertion and reason are correct but reason is not the correct explanation of assertion
(4) Both assertion and reason are correct but reason is the correct explanation of assertion



38) Identify the correct statement w.r.t given figure:

(1) In females pelvic fins bear claspers
(2) Skin is covered by cycloid scales
(3) They are mostly oviparous
(4) Mouth is located ventrally

39) Select the incorrect statement regarding arthropoda ?

(1) They have organ system level of organisation
(2) Circulatory system is open type
(3) They are pseudocoelomate animals

(4) They have jointed appendages

40) **Statement-I :** Body of *Saccoglossus* divided into anterior proboscis, collar and long trunk.
Statement-II : *Balanoglossus* show indirect development and organ system level of organization.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect
(3) Only statement-I is correct.
(4) Only statement-II is correct.

41) Match the column-I with the column-II and find out the correct answer :

	Column-I		**Column-II**
(A)	Proboscis gland	(i)	*Aurelia*
(B)	Cnidocytes	(ii)	*Psittacula*
(C)	Electric organs	(iii)	*Balanoglossus*
(D)	Air sacs	(iv)	*Torpedo*

(1) A-iii, B-i, C-ii, D-iv
(2) A-i, B-iii, C-iv, D-ii
(3) A-iii, B-i, C-iv, D-ii
(4) A-ii, B-i, C-iii, D-iv

42) There are some features of animals are given :-
Organ system level, Coelomate, Open circulation, Triploblastic, Unsegmented, Intracellular digestion.
How many characters are present in the members of phylum arthropoda -

(1) 3
(2) 4
(3) 5
(4) 2

43) What is the right sequence of structure of mollusca from outside to inside -

(1) Shell → Mantle cavity → Mantle → Visceral hump.
(2) Mantle → Shell → Mantle cavity → Visceral hump.
(3) Shell → Mantle → Mantle cavity → Visceral hump.
(4) Shell → Mantle → Visceral hump. → Mantle cavity

44) Which one of the following statement is totally wrong about the occurrence of notochord while the other three are correct ?

(1) It is present throughout life in Amphioxus
(2) It is present only in larval tail in Ascidians
(3) It is replaced by a vertebral column in adult frog

(4) It is absent throughout life in humans from the very beginning of embryological stage

45) **Statement-I :** Ctenophora show bioluminescence property and only reproduce sexually.
Statement-II : *Planaria* possess high regeneration capacity and show internal fertilization.

(1) Statement-I and II both are correct.
(2) Statement-I and II both are incorrect
(3) Only statement-I is correct.
(4) Only statement-II is correct.

46) Which of the following is correctly matched?

(1) *Funaria*: Water is required for fertilization
(2) *Adiantum*: Produce two different types of spores
(3) *Chlamydomonas*: Unicellular alga classified in kingdom Plantae by Whittaker
(4) *Volvox*: Colonial blue green algae

47) Chlorophyll a and chlorophyll b are commonly present in:

(1) Green algae
(2) Blue green algae
(3) Red algae
(4) 1 and 2 both

48) Match the Column I with Column II

Column I		Column II	
A.	*Chlorella*	P.	Naked seed plant
B.	*Equisetum*	Q.	Peat Moss
C.	*Sphagnum*	R.	Homosporous Pteridophyta
D.	*Ginkgo*	S.	Unicellular green algae

(1) A: P; B: Q; C: R; D: S
(2) A: S; B: R; C: Q; D: P
(3) A: S; B: R; C: P; D: Q
(4) A: R; B: Q; C: P; D: S

49) Male and female reproductive structures are present in different plants in:

(1) *Pinus and Cycas*
(2) *Funaria and Marchantia*
(3) *Cycas and Marchantia*
(4) *Chara and Marchantia*

50) Which of the following is not associated with all seeded plant?

(1) fruit formation
(2) Heterospory
(3) Seed formation
(4) Independent sporophyte

51) Antheridium is present in:

(1) Algae and gymnosperm
(2) Algae and Bryophyta only
(3) Algae, Bryophyta and Pteridophyta
(4) Bryophyta, Pteridophyta and Gymnosperm

52) Large size leaves with homospory are observed in:

(1) *Pteris*
(2) *Equisetum*
(3) *Selaginella*
(4) *Salvinia*

53) **Assertion (A):** *Marchantia* is a dioecious plant while *Funaria* is a monoecious plant
Reason (R): Male and female reproductive organs archegonium and antheridium respectively are present on different plant in *Marchantia* while they are on same plant in *Funaria*

(1) A and R both are true and R explain A
(2) A is true but R is false
(3) A and R both are true and R does not explain A
(4) A and R both are false

54) Pinnately compound leaves are commonly observed in:

(1) *Adiantum*
(2) *Cycas*
(3) *Adiantum* and *Cycas* both
(4) *Pinus*

55) Which of the following set of plant produce non motile gametes ?

(1) *Spirogyra, Pinus* and *Cycas*
(2) *Spirogyra, Porphyra* and *Sphagnum*
(3) *Pinus, Pisum* and *Spirogyra*
(4) *Porphyra, Nostoc* and *Pinus*

56) Endosperm of gymnosperm is:

(1) Male gametophyte
(2) Female Sporophyte

(3) Female gametophyte
(4) Megasporangia

57) Which of the following is not associated with red algae ?

(1) Pre dominance of r-phycoerythrin
(2) Fusion of gametes inside oogonium
(3) Present only in greater depth of water
(4) Some members are used as food

58) Evolutionary first plants to have embryo are:

(1) Bryophyta
(2) Pteridophyta
(3) Gymnosperm
(4) Algae

59) Antheridium and archegonium in Bryophyta and Pteridophyta are:

(1) Multicellular and unicellular respectively
(2) Multicellular and Multicellular respectively
(3) unicellular and unicellular respectively
(4) Multicellular and unicellular respectively

60) Few stages of life cycle of Pteridophyta are given below. Arrange these events in correct order starting from sex organ bearing structure and select the correct option from option given below
A: Embryo
B: Prothallus
C: Spore formation
D: Spore germination
E: Fertilization
F: Main plant

(1) B → A → C → D → E → F
(2) B → E → A → F → D → C
(3) F → C → D → B → E → A
(4) B → E → A → F → C → D

61) Two statements 1 and 2 are given below:
Statement 1 : Leaves of gymnosperms are well adapted for extremes of environment
Statement 2 : Needle like leaves of conifers reduces the surface area and reduces the transpiration loss of water

(1) Statement 1 and 2 both are false
(2) Only statement 1 is true
(3) Only Statement 2 is true
(4) Statement 1 and 2 both are true

62) Zygotes do not undergo reduction division immediately and produce a multicellular body called a sporophyte in:

(1) Bryophyta only
(2) Bryophyta and Pteridophyta only
(3) Algae, Bryophyta and Pteridophyta only
(4) Bryophyta, Pteridophyta and Gymnosperm

63) Which of the following brown algae show isogamy?

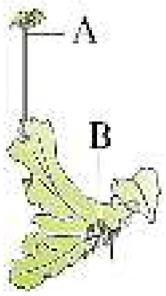
(1) *Ectocarpus*
(2) *Fucus*
(3) *Sargassum*
(4) 1 and 2 both

64) Which of the following is not associated with *Sphagnum*?

(1) It is used as packing material for trans-shipment of non-living material
(2) It has independent gametophyte
(3) Multicellular rhizoids are present
(4) It is a monoecious plant

65) Genera like *Selaginella* and *Salvinia* which produce two kinds of spores, macro (large) and micro (small) spores, are known as heterosporous. The megaspores and microspores germinate and give rise to female and male gametophytes, respectively.
Read the following statements A and B carefully and select the correct option :
Statement A : The female gametophytes in these above plants are retained on the parent sporophytes for variable periods.
Statement B : The development of the zygotes into young embryos take place within the male gametophytes.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Statement A and B both are correct
(4) Statement A and B both are incorrect

66) identify A and B in given figure : 

(1) A: Archegonia, B: Gemma cup
(2) A: Antheridium, B: Gemma cup
(3) A: Archegoniophore, B: Gemma cup

(4) A: Antheridiophore, B: Gemma cup

67) Male and female gametes in *Cycas* are

(1) Motile and non-motile respectively
(2) Non motile and motile respectively
(3) Both are motile
(4) Both are non-motile

68) Gametophytic and sporophytic both generations are independent to each other in :

(1) *Salvinia*
(2) *Pteridium*
(3) *Funaria*
(4) *Pinus*

69) A collection of species, which bears a close resemblance to one another in the morphological characters of the floral parts is known as :-

(1) Family
(2) Genus
(3) Division
(4) Variety

70) *Panthera leo, Panthera pardus and Panthera tigris* represent :-

(1) They are member of same species
(2) They are species of different genus
(3) They are different species of same genus
(4) *Panthera* is a name of species while *leo, pardus* and *tigris* represent variety

71)

Both the words in a biological name when handwritten are underlined or printed in italics to indicate :-

(1) They are endangered
(2) They are living
(3) Their Latin origin
(4) Now they are extinct

72) How many of the following terms are not related with classification of Mango ? *Mangifera,* Anacardiaceae, Poales, Dicotyledonae, Angiospermae, Monocotyledonae, Polymoniales, Convolvulaceae, Plantae

(1) One
(2) Four

(3) Three

(4) Two

73)

Consider the following four statements (a-d) and select the option which include all the correct ones only regarding Ernst Mayr :
(a) The Darwin of the 20th century.
(b) He is pioneered the currently accepted definition of a biological concept of species
(c) Mayr was awarded nobel prize in the field of Evolutionary biology

(d) His research spanned Ornithology, Taxonomy, systematics, Zoogeography and Evolution.

(1) Statements : (a) and (b) only

(2) Statements : (a), (b) and (d)

(3) Statements : (b), (c) and (d)

(4) Statements : (a), (b), (c) and (d)

74) Study the given statements carefully and give the answer :-
(A) A genus is a group of related species which has more characters in common in comparison to species of other genera
(B) Families are characterised on the basis of both vegetative and reproductive features of plants species.
(C) Felidae and canidae are the families of cat and dog respectively
(D) To determine relations in higher taxa is more difficult in comparison to lower taxa.
How many statements are correct ?

(1) One

(2) Four

(3) Three

(4) Two

75) What is incorrect for chemosynthetic autotrophic bacteria :-

(1) Oxidise various organic and inorganic substances

(2) Use the energy released through oxidation for their ATP production

(3) Play great role in recycling nutrients like nitrogen, phosphorus, iron and sulphur

(4) They are most abundant in nature

76) Study the following statements :
(I) All prokaryotic organisms were grouped together under Kingdom fungi.
(II) The unicellular eukaryotic organisms were placed in Kingdom protista.
(III) *Chlorella* and *Chlamydomonas* both have cell wall.
(IV) *Paramoecium* and *Amoeba* lack cell walls.
(V) Kingdom protista has brought together *Chlamydomonas* , *Chlorella* with *Paramoecium* and *Amoeba*.
Which of the statements given above are correct according to Whittaker classification ?

(1) II, III and IV only

(2) II, III, IV and V only
(3) I, II, III and IV only
(4) I, II, III, IV and V

77) How many of the following fungi have septate and Branched Mycelium with absence of Asexual spores
Neurospora, Alternaria, Claviceps, Agaricus, Colletotrichum, Penicillium, Trichoderma, Aspergillus, Mucor, Rhizopus and Albugo

(1) 8
(2) 3
(3) 1
(4) All

78) A proteinaceous covering which is elastic enough to enable turning and flexing of the cell, yet rigid enough to prevent excessive alternation in shape is :-

(1) Pellicle in *Euglena*
(2) Silica in diatoms
(3) Budding in yeast
(4) Plasma membrane in *Gonyaulax*

79) Identify the correct match from column-I and II :

	Column I		**Column II**
(i)	Euglena	(a)	Saprophytic
(ii)	Diatom	(b)	Diatomaceous earth
(iii)	Slime mould	(c)	Red tide
(iv)	Dinoflagellates	(d)	*Gonyaulax*
		(e)	Pellicle
		(f)	Chief producer of oceans

(1) (i)-e, (ii)-b,f (iii)-c, (iv)-a,d
(2) (i)-e, (ii)-b,f (iii)-a, (iv)-c,d
(3) (i)-e, (ii)-b,c,f (iii)-a, (iv)-d
(4) (i)-e, (ii)-b,c,f (iii)-d, (iv)-a

80) Select the wrong statement :

(1) Diatoms are photosynthetic
(2) Diatoms are microscopic and float passively in water
(3) The walls of diatoms are easily destructible
(4) 'Diatomaceous earth' is formed by the cell walls of diatoms.

81) **Assertion :** Mesosome increases the surface area of plasma membrane for enzymatic content.

Reason : Mesosomes are foldings of plasma membrane towards cell wall.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

82) Which one of the following is exclusively found in blue green algae but not found in any other prokaryotes :-

(1) Peptidoglycan's cell wall
(2) Chlorophyll 'a'
(3) Lipoproteins cell membrane
(4) Nitrogen fixation ability

83) **Statement -I :** Mycoplasma is smallest living organism that can survive without oxygen.
Statement -II : Members of protista are primary aquatic.

(1) Both the statements are correct.
(2) Both the statements are incorrect.
(3) Statement-I is correct whereas statement-II is incorrect.
(4) Statement-I is incorrect whereas statement-II is correct.

84) **Assertion :-** Archaebacteria live in extremely adverse conditions such as very high temperature and high salt concentration.
Reason :- Archaebacteria differ from other bacteria in having a different cell wall structure and complex cell membrane.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

85) According to five kingdom system scheme, which statement is common about given organism :-
Euglena, Nostoc, Chlorella, Diatoms and *Dinoflagellates*

(1) All are unicellular eukaryotes
(2) All are multicellular eukaryotic organism
(3) All are photosynthetic unicellular prokaryotes
(4) They are photosynthetic and act as producer in nature

86)

Select the option which includes all the correct ones only :-
(A) Diatoms → Chief producer of ocean
(B) Dinoflagellate → Without flagella
(C) Slime mould → Decomposer nature

(D) Euglenoids → Sometime behave like predator

(E) Protozoa → Unicellular prokaryotes

(1) A, B and C

(2) A, B, C and D

(3) B, C, D and E

(4) A, C and D

87) Read the following statement (A-D) (A) Cell wall of Fungi consists of chitin.
(B) Fungi are heterotrophic
(C) Fungi can also live as symbionts in association with algae as lichens and with roots of higher plants as mycorrhiza
(D) Fusion of two nuclei called plasmogamy
How many of the above statement are correct ?

(1) Two

(2) Three

(3) Four

(4) One

88) Read the given below statements and choose correct answer -
Statement-I : *Albugo* do not show karyogamy immediately after plasmogamy. In *Albugo* an intervening dikaryotic stage (n + n, i.e. two nuclei per cell) occurs : such a condition is called a dikaryon and the phase is called dikaryophase.
Statement-II : *Neurospora* and *Puccinia*, show karyogamy immediately after plasmogamy.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct and statement II is incorrect.

(4) Statement I is incorrect and statement II is correct.

89) Choose the incorrect pair of following :-

(1) Phycomycetes-aquatic habitat

(2) Ascomycetes-Rust and smut disease

(3) Basidiomycetes-lack sex organs

(4) Deuteromycetes-help in mineral cycling

90) Select the correct pair :-

(1) Lichens – Grow in Polluted areas.

(2) Prions – Abnormally folded protein

(3) Virus – Cause Potato spindle tuber disease

(4) Viroids – Consist of protein and nucleic acid

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**2**	**3**	**3**	**4**	**4**	**2**	**1**	**3**	**4**	**3**	**2**	**3**	**2**	**2**	**1**	**2**	**3**	**2**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**3**	**4**	**1**	**3**	**1**	**3**	**2**	**1**	**1**	**4**	**3**	**4**	**1**	**2**	**3**	**4**	**3**	**3**	**1**
Q.	41	42	43	44	45															
A.	**2**	**3**	**1**	**3**	**2**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**1**	**3**	**4**	**4**	**3**	**1**	**2**	**4**	**3**	**3**	**3**	**4**	**3**	**3**	**3**	**3**	**3**	**1**	**4**	**1**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**3**	**3**	**3**	**3**	**2**	**3**	**4**	**1**	**3**	**1**	**3**	**2**	**2**	**4**	**3**	**4**	**1**	**4**	**4**	**1**
Q.	86	87	88	89	90															
A.	**2**	**3**	**4**	**3**	**2**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**3**	**3**	**3**	**2**	**1**	**4**	**3**	**1**	**1**	**1**	**1**	**2**	**4**	**3**	**2**	**1**	**4**	**1**	**4**	**1**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**2**	**1**	**4**	**4**	**3**	**3**	**3**	**3**	**4**	**2**	**2**	**2**	**1**	**1**	**4**	**2**	**3**	**4**	**3**	**1**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**3**	**2**	**3**	**4**	**1**	**1**	**1**	**2**	**3**	**1**	**3**	**1**	**2**	**3**	**3**	**3**	**3**	**1**	**2**	**4**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**4**	**4**	**1**	**1**	**1**	**4**	**1**	**2**	**2**	**3**	**3**	**2**	**2**	**2**	**4**	**2**	**3**	**1**	**2**	**3**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**3**	**2**	**1**	**1**	**4**	**4**	**2**	**2**	**2**	**2**										

SOLUTIONS

PHYSICS

1) $A\cos\theta = \dfrac{\vec{A}.\vec{B}}{B} = \dfrac{(\hat{i}-\hat{j}).(-\hat{i}+\hat{j})}{\sqrt{2}} = -\sqrt{2}$

2)

$\tan\theta = \dfrac{y}{x} = \dfrac{1}{1} = 1 \Rightarrow \theta = 45°$

3) $|\hat{a}| = |\hat{b}| = 1, |\hat{a}+\hat{b}| = \sqrt{3}$

$\Rightarrow |\hat{a}|^2 + |\hat{b}|^2 + 2|\hat{a}||\hat{b}|\cos\theta = 3$

$\Rightarrow \cos\theta = \dfrac{1}{2}$

$\Rightarrow \theta = 60°$(Angle between $\hat{a}$ and $\hat{b}$)

Now,

$(2\hat{a}-5\hat{b}).(3\hat{a}+\hat{b})$

$= 6\hat{a}.\hat{a} - 15\hat{b}.\hat{a} + 2\hat{a}.\hat{b} - 5\hat{b}.\hat{b}$

$= 1 - 13\ \hat{a}.\hat{b}$

$= 1 - 13\ (1.1.\cos 60°)$

$= 1 - \dfrac{13}{2} = -\dfrac{11}{2}$

4) $\dfrac{2}{-4} = \dfrac{3}{-6} = -\dfrac{1}{\lambda}$

$\lambda = 2$

5) $\hat{n} = \pm\left(\dfrac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}\right)$

$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} = -3\hat{k}$

$\hat{n} = \pm\dfrac{-3\hat{k}}{3} = \mp\hat{k}$ Ans.

6) $\vec{r} = (\sin t\,\hat{i} + \cos t\,\hat{j} + t\,\hat{k})m$

$\vec{v} = \dfrac{d\vec{r}}{dt} = (\cos t\,\hat{i} - \sin t\,\hat{j})$

$\vec{a} = \dfrac{d\vec{v}}{dt} = (-\sin t\,\hat{i} - \cos t\,\hat{j})$

According to question

$\vec{r}\cdot\vec{a} = 0$

$\Rightarrow \quad -\sin^2 t - \cos^2 t = 0$

$\Rightarrow \quad 1 = 0$ which is not possible

7)

∴ $x = 8t - 4t^2$

at $t = 0$, $x = 0$

∴ $v = \frac{dx}{dt} = 8 - 8t$ if $v = 0$, $t = 1$

at $t = 1$, $x = 8 - 4 = 4$m

at $t = 2$, $x = 8(2) - 4(2)^2 = 0$

so, travelled distance $= |(4-0)| + |(0 - 4)|$

$= 4 + 4 = 8$m

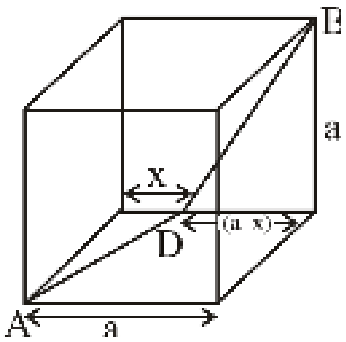
8)

$s = kt^3$ (k = constant)

∴ $v = \frac{ds}{dt} = 3kt^2$

∴ $a = \frac{dv}{dt} = 6kt \quad \therefore a \propto t$

9)



$t \propto$ distance

Total distance = AD + DB

$= \sqrt{a^2 + x^2} + \sqrt{a^2 + (a-x)^2}$

For min. dist. $\frac{d}{dx}(AD + DB)$

$= \frac{1}{2\sqrt{a^2 + x^2}} \times 2x + \frac{2(a-x)(-1)}{2\sqrt{a^2 + (a-x)^2}} = 0$

$\Rightarrow 2x + (2a - 2x)(-1) = 0$

$\Rightarrow 4x = 2a$

$\Rightarrow x = a/2$

∴ min dist. $= \sqrt{a^2 + \left(\frac{a}{2}\right)^2} + \sqrt{a^2 + \left(\frac{a}{2}\right)^2}$

$= \sqrt{5}\,a$

∴ min time $= \frac{\sqrt{5}a}{u}$

10) $\vec{v} = \vec{u} + \vec{a}t$
$\vec{v} = 3\hat{i} + 4\hat{j} + (0.4\hat{i} + 0.3\hat{j})$
$= 7\hat{i} + 7\hat{j}$
$|\vec{v}| = 7\sqrt{2}$ units

11) **Explanation :** A body slides down an inclined plane with height h, length l, and angle of inclination θ. We need to find the time it takes for the body to travel from the top to the bottom of the incline.

Concept : motion of body of incline plane.

Formula : Acceleration along the Incline = g sin(θ), where g is the acceleration due to gravity.

Distance traveled (I) = $(1/2)at^2$ (assuming initial velocity is zero)

Relationship between h, l, and θ: sin(θ) = h/l

Calculation :
$\sin\theta = \frac{h}{t} \Rightarrow 1 = \frac{h}{\sin\theta}$
$s = u\,t + \frac{1}{2}at^2$
$\frac{h}{\sin\theta} = 0 + \frac{1}{2}(g \sin\theta)\,t^2$
$t = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}}$
Hence, the correct answer is option (3).

12)

∴ $S_A = S_B$ ∴ $40t = 0 + \frac{1}{2} \times 4t^2$
∴ $t^2 - 20t = 0$ ∴ $t = 20$ sec.

13) $4 = \frac{(200)^2}{2a}$
$9 = \frac{(v)^2}{2a}$
$\frac{4}{9} = \frac{4 \times 10^4}{v^2}$
v = 300 cm/s

14) $S_{rel} = u_{rel}t + \frac{1}{2} a_{rel}t^2$

$100 + 125 = (10 + 10)\ t + \frac{1}{2} (0.3 + 0.2)\ t^2$

$900 = 80\ t + t^2$

$\Rightarrow\ t^2 + 90\ t - 10\ t - 900 = 0$

$t = -90,\ t = 10$ sec

15)

$V_x = 4t$ $\quad$ $v_y = 3$

$\int_0^x dx = 4\int_0^2 t\,dt$ $\quad$ $\int_2^y dy = 3\int_0^2 dt$

$x = 4\left[\frac{t^2}{2}\right]_0^2$ $\quad$ $y - 2 = 3[t]_0^2$

$x = 4 \times 2 = 8$ $\quad$ $y = 8$

So, at t = 2sec, $\vec{r} = 8\hat{i} + 8\hat{j}$

16) $H_1 = \frac{u^2\sin^2\theta}{2g}$

and $H_2 = \frac{u^2\sin^2(90-\theta)}{2g} = \frac{u^2\cos^2\theta}{2g}$

$H_1H_2 = \frac{u^2\sin^2\theta}{2g} \times \frac{u^2\cos^2\theta}{2g} = \frac{(u^2\sin 2\theta)^2}{16g^2} = \frac{R^2}{16}$

$\therefore R = 4\sqrt{H_1H_2}$

17) $t = \frac{d}{v}\ \ \therefore 5 = \frac{60}{v}\ \ \therefore v = 12\,m/s$

 $v_m = \sqrt{v^2 + v_R^2} = \sqrt{(12)^2 + (5)^2} = \sqrt{169}$

= 13 m/s

18)

Time to meet the cars : $t = \frac{d}{v_1 + v_2}$

Distance travelled by bird in this time

$s = v_3t = \frac{v_3d}{v_1 + v_2} = \frac{10 \times 2000}{(20 + 30)} = 400m$

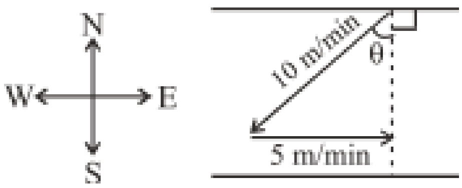
19) Relative velocity of bird w.r.t
train = 25 + 5 = 30 m/s
time taken by the bird to cross the train

$t = \frac{210}{30} = 7\,\text{sec}$

20) $\vec{v} = a\,\hat{i} + b\,\hat{j} \Rightarrow u\cos\theta = a$ and $u\sin\theta = b$

$R = 2H_{max}$ $\tan\theta = \frac{b}{a}$ also $\frac{u^2\sin 2\theta}{g} = \frac{2u^2\sin^2\theta}{2g} \Rightarrow \tan\theta = 2 \therefore \frac{b}{a} = 2 \Rightarrow b = 2a$

21) $\vec{v}_{BW} = \vec{v}_{BG} - \vec{v}_{WG} = 6\hat{i} + 8\hat{j}$

22) 

$\sin\theta = \frac{5}{10} = \frac{1}{2}$

$\theta = 30°$

Angle with downstream

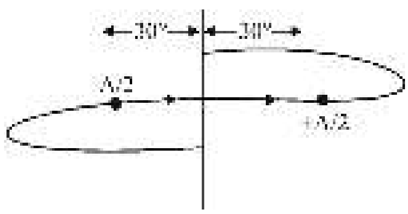
$= 90° + 30° = 120°$

23)

$x = A\sin\omega t,$

$v = A\omega\cos\omega t,$

$a = -A\omega^2\sin\omega t$

24) 

Time period of SHM $= \frac{2\pi}{\omega}$

∴ Time for traveling 60° is $= \frac{2\pi}{\omega}\frac{60}{360}$

$= \frac{\pi}{3\omega}$

25) velocity, $v = \omega\sqrt{a^2 - (a/2)^2} = \frac{\sqrt{3}}{2}a\omega$

Acceleration, $f = \frac{\omega^2 a}{2}$

∴ v = f (given)

∴ $\frac{\omega^2 a}{2} = \frac{\sqrt{3}}{2}a\omega$

$2\pi n = \sqrt{3}$

$n = \frac{\sqrt{3}}{2\pi}$

26) $a = \frac{d\nu}{dt}$

27) $T = 2\pi\sqrt{\frac{m}{k}}$

$T' = \frac{T}{\sqrt{3}} \Rightarrow k_{eff} = 3k$

28)

Given, KE = PE

$\frac{1}{2}K(A^2 - x^2) = \frac{1}{2}Kx^2$

$x = \frac{A}{\sqrt{2}}$

At t = 0, x = 0

For $x = A\sin\omega t = \frac{A}{\sqrt{2}}$

$\omega t = \frac{\pi}{4}, t = \frac{\pi}{4\omega} = \frac{\pi}{4}\left(\frac{T}{2\pi}\right)$

$\Rightarrow \quad t = \frac{T}{8} = \frac{8}{8} = 1$ sec

29) $\frac{1}{2}kA^2 = U_{max} \Rightarrow k = \frac{2U_{max}}{A^2} = \frac{2 \times 24}{(2)^2} = 12\,N/m$

30)

k = 200 N/m

$mg > m\omega^2 A$

$A = \frac{g}{\omega^2} \qquad \omega^2 = \frac{k}{m}$

$A = \frac{g}{k} \times m$

$A = \frac{10}{200} \times 2$

$A = \frac{1}{10}m$

A = 10 cm

31)

Using, $x = A\sin(\omega t + \phi)$

at t = 2 sec → x = 0

$0 = A\sin\left(\frac{2\pi}{16} \times 2 + \phi\right)$

$0 = \sin\left(\frac{\pi}{4} + \phi\right)$

$\phi = -\frac{\pi}{4}$

So, $x = A\sin\left(\frac{\pi}{8}t - \frac{\pi}{4}\right)$

$v = \frac{\pi}{8} A\cos\left(\frac{\pi}{8}t - \frac{\pi}{4}\right)$

at t = 4 sec ⇒ v = 4 m/s

$4 = \frac{\pi}{8} A\cos\left(\frac{\pi}{8} \times 4 - \frac{\pi}{4}\right)$

$4 = \frac{\pi A}{8} \times \frac{1}{\sqrt{2}} \Rightarrow A = \frac{32\sqrt{2}}{\pi}$ (in m)

32)

$U = 2 - 20x + 5x^2$

$F = -\frac{dU}{dx} = -(-20 + 10x)$

$F = 20 - 10x$

x = 2 (M.P.)

x=–3 x=2 x=7

A = 5 cm

$x_{max} = 7$ cm

33) $V = \sqrt{\frac{T}{\mu}} = \frac{\omega}{k}$ μ → linear mass density

$\Rightarrow T = \frac{\omega^2 \mu}{k^2} = \left(\frac{420}{21}\right)^2 \times 0.2$

= 80 N

34) $k = \frac{2\pi}{\lambda} = \frac{2\pi}{4/5} = 7.85$

$v = \frac{\omega}{k} \Rightarrow \omega = vk = 128 \times 7.85 = 1005$

and wave moves in + x direction

35) Velocity $\propto \sqrt{T}$

36)

$7I_0 = I_0 + 9I_0 + 2 \times I_0 \times 3 \cdot \cos\Delta\phi$

$-3I_0 = 6I_0 \cdot \cos\Delta\phi$

$\cos\Delta\phi = -\frac{1}{2} = \cos 120°$

37)

Speed of sound

$\upsilon = \sqrt{\left\{\frac{\gamma RT}{M}\right\}} \propto \sqrt{T}$ ∴ $\frac{\upsilon_2}{\upsilon_1} = \sqrt{\frac{T_2}{T_1}}$

Given, $\frac{\upsilon_2}{\upsilon_1} = 3$ ∴ $3 = \sqrt{\frac{T_2}{T_1}}$ or $\frac{T_2}{T_1} = 9$

$\Rightarrow T_2 = 9T_1$

Here : $T_1 = 0°C = 273$ K

∴ $T_2 = 9 \times 273$ K

= 2457 K

= (2457 – 273)°C

= 2184°C

38)

Time period of simple pendulum depends on centre of mass position

39)

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$, where $\Delta\phi = \frac{2\pi}{\lambda}\Delta x$

40)

y = 5x2 – 2x + 1

$\frac{dy}{dx} = 10x - 2$ $\quad$ 10x – 2 = 0 $\quad x = \frac{1}{5}$

$\frac{d^2y}{dx^2} = 10$ $\quad y_{min} = 5\left(\frac{1}{5}\right)^2 - 2\left(\frac{1}{5}\right) + 1 = \frac{4}{5}$

41) $\int_0^{x_1} y\,dx = 15$ = Area under curve from 0 to x_1

= Area of triangle

$15 = \frac{1}{2}(x_1)(3) \Rightarrow \boxed{x_1 = 10}$

42) At x-axis , y = 0 so $x^2 + 2 - 3x = 0 \Rightarrow x = 1$ and 2

Therefore curve intersects the x-axis at (1, 0) and (2, 0).

43) As tan θ is decreasing slope is decreasing

44) $(1.002)^3 = (1 + 0.002)^3 = 1 + .006$
$= 1.006$

45)

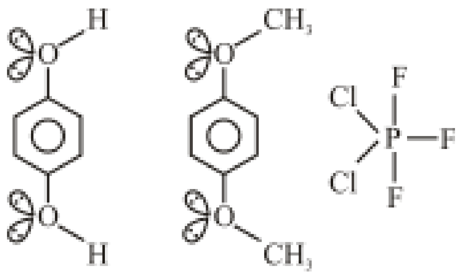
12 cm

CHEMISTRY

46) SF_6 = Octahedral all six S – F bond length are same

47) F_2 is diamagnetic

48) B.O. ↑ $O_2 \Rightarrow 2$ $O_2^+ = 2.5$



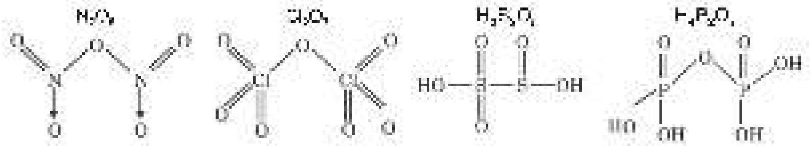
49) $\mu \neq 0$ $\mu \neq 0$ $\mu \neq 0$

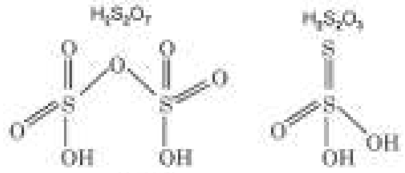
50)

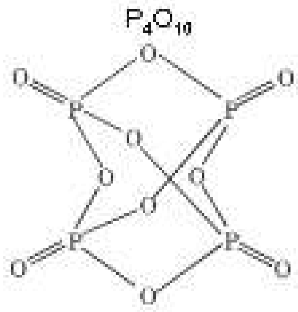
	Shape
NO_2^+	Linear
NO_2^-	V-shape
PCl_5	TBP
BrF_5	Square pyramidal

XeF_4, ICl_4^- Square planar XeO_4 Tetrahedral
$TeCl_4$ Bent T-shape

51) ⇒ O_2 is paramagnetic (Using M.O.T.)
⇒ In KO_2, O_2^- (Superoxide ion) is present
O_2^- is paramagnetic

52) 





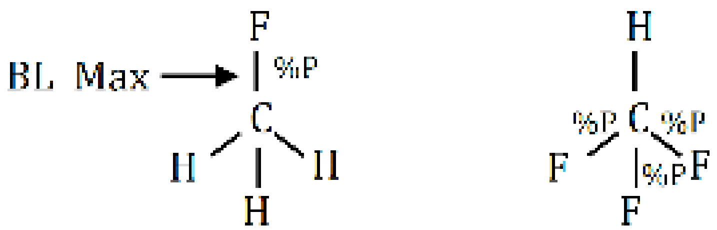
53) $[PCl_4]^+ [PCl_6]^-$

↓

$sp^3d^2[s + p_x + p_y + p_z + d_{z^2} + d_{x^2-y^2}]$

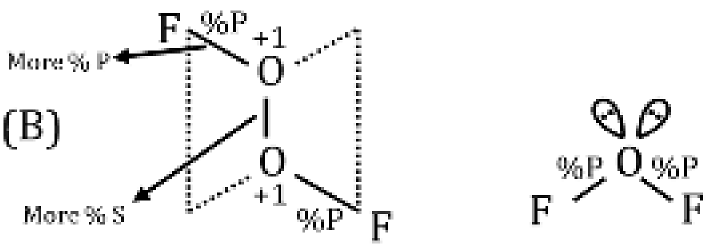
54)

In IF_7 5 bond angle of 72°, 10 bond angle of 90°

(A) 

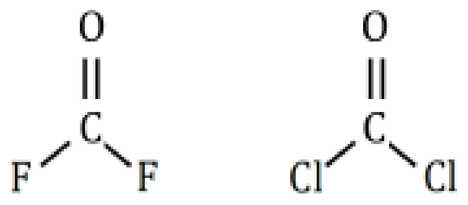
55)

$CH_3F > CHF_3$

(B) 

$BL_{O-F} \Rightarrow O_2F_2 > OF_2$

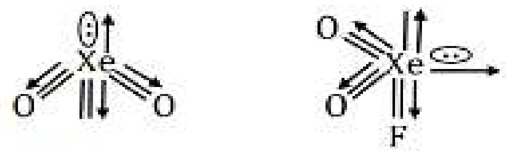
(C)



Cl is Bulky Group

$\therefore$ BA $\Rightarrow$ $OCF_2 < OCCl_2$

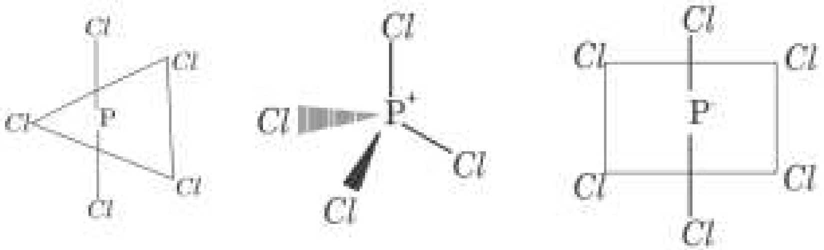
56)



$\mu \neq 0$ Polar

$\mu \neq 0$ Polar

57)



sp^3d TBP

sp^3 Tetrahedral

sp^3d^2 Octahedral

58)

$:NH_3$ has max. s-character as B.A.T.

59) **Answer - Option(3)**

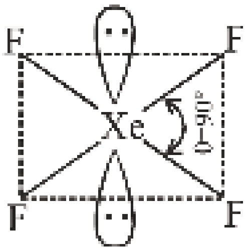
Explanation - Let's analyze each given order and identify the incorrect one.

Concept - The incorrect order is 3. Bond strength order : $F_2 > Cl_2 > Br_2 > I_2$

While F_2 has a *high* bond strength, it's *not* the highest among the halogens. The actual bond strength order is:

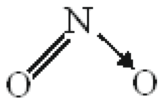
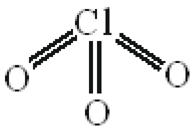
$Cl_2 > Br_2 > F_2 > I_2$

The reason F_2 is an exception is due to its small size. The lone pairs of electrons on the fluorine atoms experience significant repulsion, weakening the F-F bond. While fluorine has a high electronegativity (which usually leads to stronger bonds), the interelectronic repulsion outweighs this effect in the case of F_2.

60) 

Bond angle indicated will be equal to 90° as the structure is symmtrical.

61)

(A) $CH_3 \rightarrow sp^2$
(B) $CF_3 \rightarrow sp^3$
(C) NO_2  sp^2
(D) Cl_3  sp^3

62)

(A) → due to absence of 'd' orbital in nitrogen
(B) → P –Cl length > P – F length because Cl atoms size is large
(C) $PCl_5 \rightarrow sp^3d^2$
(D) ICl_7 not exist due to steric hindrance

63)

IF_7 – non-planer and non polar
SO_2 – planer and polar
SF_4 – non-planer and polar
CS_2 – planer and non polar

64)

According to VSEPRT shape of NH_3 and AsH_3 is pyramidal and shape of H_2O is V/bent shape for which hybridization is not necessary.

65)

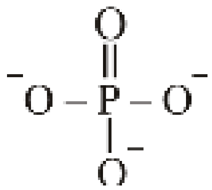
Bond angle order : $CH_3 > CH_4$
Order of %s character in Xe-O bond : $XeO_4 > XeO_3$

66)

sp^3 hybridisation can only have pπ-dπ bond as it can not form pπ-pπ bond.

67)

If hybridization is not considered in CH_4 molecule then shape of CH_4 is uncertain.

68) 

Average FC on O atom = $\dfrac{-1-1-1+0}{4}$

$= -\dfrac{3}{4}$

69)

O_2F_2 is polar and non planer

70)

In C_2H_4 C – C bond order is 2 and in C_2F_6 C – C bond order is 1.

71) $xA + yB \rightarrow zC$

$-\dfrac{1}{x}\dfrac{d[A]}{dt} = -\dfrac{1}{y}\dfrac{d[B]}{dt} = \dfrac{1}{z}\dfrac{d[c]}{dt}$

x : y : z

$1 : 1 : \dfrac{2}{3}$

$\Rightarrow 3 : 3 : 2$

72)

at 50°C, $K_1 = 1.5 \times 10^7$

at 100°C, $K_2 = 4.5 \times 10^7$

from Arrhenius equation

$\log \dfrac{K_2}{K_1} = \dfrac{E_a}{2.303R}\left(\dfrac{T_2 - T_1}{T_1 T_2}\right)$

$\log_{10} \dfrac{4.5 \times 10^7}{1.5 \times 10^7} = \dfrac{E_a}{2.303 \times 8.314}\left(\dfrac{373 - 323}{373 \times 323}\right)$ $E_a = \dfrac{\log 3 \times 2.303 \times 8.314 \times 373 \times 323}{50}$

$= 2.2 \times 10^4$ J/mole

73) Rate always depends upon temperature.

74) $r = K[P][Q_2]$ (rds)

From (i) step

$K_c = \frac{[P]^2}{[P_2]}$

$[P] = K_c^{\frac{1}{2}}[P_2]^{\frac{1}{2}}$

$r = kK_c^{\frac{1}{2}}[P_2]^{\frac{1}{2}}[Q_2]$

order $= \frac{3}{2}$

75) $\frac{0.08}{0.04} = \left(\frac{0.2}{0.1}\right)^a \left(\frac{0.4}{0.4}\right)^b$

a = 1

$\frac{0.08}{0.04} = \left(\frac{0.2}{0.2}\right)^a \left(\frac{0.4}{0.2}\right)^b$

b = 1

76)

$t_{1/2} \propto (a)^{1-n}$

$\left(\frac{3.64}{1.82}\right) = \left(\frac{0.5}{1.0}\right)^{1-n}$

$(2) = \left(\frac{1}{2}\right)^{1-n}$

1 – n = – 1

n = 2

77) $NH_4NO_2(aq) \rightarrow N_2(g) + 2H_2O(\ell)$ ni a 0 0

nt a – x x 2x

n_∞ 0 a 2a

a = 70 & x = 40

a – x = 30

$k = \frac{2.303}{1200}\log\frac{70}{30}\sec^{-1}$

$= \frac{2.303}{1200}\log\frac{7}{3}\sec^{-1}$

78)

Solution :-

Matching (Short Reasoning):

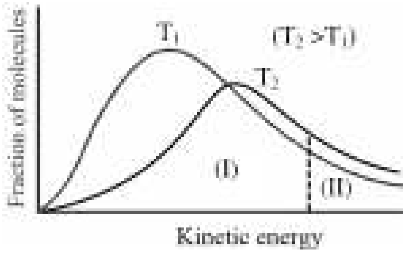
• (A) [B] vs t → (P): Product concentration [B] increases from zero and levels off as reactant A is consumed ($[B] = [A]_0(1 - e^{-kt})$). This is a **growth curve (P)**.

• (B) Rate vs [A] → (Q): For a first-order reaction, Rate = k[A]. This is a **straight line through the origin with a positive slope (Q)**.

• (C) [A] vs t → (R) : Reactant concentration [A] decreases exponentially with time ($[A] = [A]_0^{e^-}kt$). This is an **exponential decay curve (R)**.

• (D) ln[A] vs t → (S) : The integrated rate law is $lm[A] = \ln[A]_0 - kt$. This is a **straight line with a negative slope (S)**.

Final Answer :- (2)

79) 

80) **Solution/Explanation/Calculation:**
Given data
$K = .001\ Ms^{-1}$
Buy unit of K, It is a zero order reaction.
→ Formula
$A_t = A_0 - kt$
or $A_t = 1 - 0.001 \times 600$
or $A_t = 1 - 0.6 = 0.4$ M
and $B_t = A_0 - A_t = 1 - 0.4 = 0.6$ M
Hence, option (3) is correct.

81) **Explanation & Solution :**

A. **Endothermic Reaction:** The products have higher potential energy than the reactants.
B. **Rate-Determining Step:** The step with the highest activation energy (the largest peak in the diagram) determines the overall rate of the reaction.

82)

$10^{16}e^{-2000/T} = 10^{14}e^{-500/T}$
$100 = e^{1500/T}$
$T = \dfrac{1500}{2 \times 2.303} = 325.66$ K

83) **Mathematical calculation**
Let initial [x] = a, change = –x
Then, [x] = a – x, [y] = 2x
At time t,
⇒ a – x = 2x ⇒ a = 3x ⇒ x = a⁄3
 [x] = a – x = a – a⁄3 = (2a⁄3)This is when concentration becomes 2⁄3 of initial value
⇒ $t = t_{1/3}$
Final Answer : option (4)

84) Unit of $K = \left[\dfrac{\text{mol}}{\text{lit.}}\right]^{1-n}$ time^{-1}
(A) for zero order = mol L^{-1} S^{-1}
(B) for second order = mol^{-1} L S^{-1}

(C) for $3/2^{th}$ order = $mol^{-1/2}\ L^{1/2}\ S^{-1}$
(D) for 1^{st} order = S^{-1}

85)

Solution:
(A) Radioactive decay: The rate of decay is proportional to the number of undecayed nuclei. Thus, it is a (Q) First order reaction. (Rate $\propto [N]$)
(B) Decomposition of HI on gold surface at high pressure: At high pressure, the gold surface (catalyst) becomes saturated with HI . The reaction rate becomes independent of HI concentration. Thus, it is a (R) Zero order reaction. (Rate = k)
(C) Saponification (e.g., of an ester like $CH_3COOC_2H_5$ with NaOH): The rate depends on the concentration of both the ester and the alkali. Rate = $k[CH_3COOC_2H_5][OH^-]$.
Thus, it is typically a (S) Second order reaction.
(D) Inversion of cane sugar in excess of water: The reaction is $C_{12}H_{22}O_{11}+ H_2O \rightarrow C_6H_{12}O_6$ (glucose) $+C_6H_{12}O_6$ (fructose). Since water is in large excess, its concentration remains practically constant.
Rate $= k'[C_{12}H_{22}O_{11}]$. Thus, it is a (P) Pseudo first order reaction.

86) NCERT (OLD) XII Part - I Pg.No.98

$$r_{instantaneous} = \lim_{\Delta t \to 0}\left(\pm\frac{\Delta C}{\Delta t}\right) = \pm\frac{dc}{dt}$$

87)

$t_{99\%} = 32$ min
$t_{99\%} = 2\ t_{90\%}$
$t_{99.9\%} = 3\ t_{90\%}$
$\frac{t_{99\%}}{t_{99.9\%}} = \frac{2}{3}$
$t_{99.9\%} = \frac{3}{2} \times 32 = 48$ min

88)

Reaction: $2SO_2 + O_2 \rightarrow 2SO_3$
SO_3 forms at 100 g/min → corresponds to 2 mol parts
So O_2 reacts in 1 mol part
□ Rate of O_2 disappearance = (1/2) × 100 = 50 g/min
BUT O_2 molar mass = 32, SO_3 = 80
So actual mol rate:
100 g SO_3/min ÷ 80 = 1.25 mol/min
O_2 disappears at 1/2 × 1.25 = 0.625 mol/min
Mass of O_2 disappearing = 0.625 × 32 = 20 g/min

89)

For elementary reaction order is equal to stoichiometric coefficients.

90) $$\frac{\text{Rate}_{\text{Ist}}}{\text{Rate}_{\text{zero}}} = \frac{\left(\frac{0.693}{t_{1/2}}\right) \times 1}{\left(\frac{1}{2 \times t_{1/2}}\right)} = 2 \times 0.693$$

BIOLOGY

91)

Old NCERT, Pg. # 115

92)

Old NCERT, Pg. # 113

93)

Old NCERT, Pg. # 114

94)

Old NCERT, Pg. # 112

95)

Old NCERT, Pg. # 111

96)

Old NCERT, Pg. # 112

97) NCERT (XI) Pg. # 114

98)

NCERT (XI^{th}) Pg. # 113

99) NCERT XI^{th} P.No. 113

100) Pg. No. 116 NCERT 2022 - 2023 Edition

101) **Diaphragm:** Frogs do not have a diaphragm for respiration. They breathe through their skin and lungs.
Anus: Frogs have a cloaca, a single opening for the digestive, urinary, and reproductive systems.
Tail: Frogs do not have tails in their adult stage. Cortex, medulla and pyramid absent in kidney of frog ans-1

102)

New NCERT, Pg. # 83

103)

New NCERT, Pg. # 80

104)

New NCERT, Pg. # 83

105)

New NCERT, Pg. # 82

106)

NCERT Pg.#118

107)

Old NCERT, Pg. # 101

108)

Old NCERT, Pg. # 102

109)

Old NCERT, Pg. # 101

110)

Old NCERT, Pg. # 103

111)

Old NCERT, Pg. # 104

112)

Old NCERT, Pg. # 101

113)

Old NCERT, Pg. # 102, 103

114)

NCERT-XI, Pg. No. # 44

115)

The correct answer is **3 Nereis.**

A. **Dioecious** means that a species has separate male and female individuals.
B. **Nereis** (a type of marine worm) are dioecious, meaning they have distinct male and female individuals.

116)

The correct answer is Option 3: Echinodermata

A. Echinoderms exhibit radial symmetry as adults. This means their bodies can be divided into similar halves along multiple planes passing through the central axis. Examples include starfish and sea urchins.
B. However, the larvae of echinoderms have bilateral symmetry. This indicates their evolutionary relationship to other bilaterally symmetrical animals.

117)

NCERT-XI, Pg. No. # 42

118)

NCERT-XI, Pg. No. # 39

119)

The correct answer is Option 4: lchthyophis

A. lchthyophis belongs to the order Gymnophiona, which are commonly known as caecilians.
B. Caecilians are limbless amphibians, meaning they lack legs.

120)

NCERT-XI, Pg. No. # 46, 47

121)

NCERT-XI Pg. # 47, 48

122)

NCERT Pg. # 41

123)

NCERT-XI, Pg. No. # 38

124)

NCERT-XI, Pg. No. # 49, 50

125)

NCERT-XI, Pg. No. # 44

126)

NCERT XI, Pg # 53 to 55

127) NCERT, Pg # 46, 48

128)

NCERT-XI, Pg. No. # 47

129) NCERT, Pg # 44

130) NCERT Pg. # 54

131)

NCERT-XI, Pg. No. # 41, 45, 48

132) NCERT, Pg.# 44

133) NCERT Pg # 54

134)

NCERT-XI, Pg. No. # 46

135) NCERT Pg. # 51

136) NCERT-XIth, Pg. # 13, 24, 29, 32

137) NCERT-XIth, Pg. # 26

138) NCERT-XIth, Pg. # 26,30,32, 33

139) NCERT-XIth, Pg. # 29, 33

140) NCERT-XIth, Pg. # 33

141) NCERT-XIth, Pg. # 29, 32

142) NCERT-XIth, Pg. # 30, 32

143) NCERT-XIth, Pg. # 29, 30

144) NCERT-XIth, Pg. # 30, 32

145) NCERT-XIth, Pg. # 24,27,33

146) NCERT-XIth, Pg. # Module 1

147) NCERT-XIth, Pg. # 26,27

148) NCERT-XIth, Pg. # 29

149) NCERT-XIth, Pg. # 29, 32

150) NCERT-XIth, Pg. # 32

151) NCERT-XIth, Pg. # 33

152) NCERT-XIth, Pg. # 29, 32, 33

153) NCERT-XIth, Pg. # 24, 26

154) NCERT-XIth, Pg. # 29, 30

155) NCERT (XII) ; Page No. # 32

156) NCERT-XIth, Pg. # 28

157) NCERT-XIth, Pg. # Module 1

158) NCERT-XIth, Pg. # 32

159) NEW NCERT-XI, Pg # 7

160)

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161)

NCERT (XI) Pg. # 4

162)

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163) NCERT XI Pg # 2

164) NCERT Pg. # 7, 8

165) NCERT XI Pg. # 13

166) NCERT XI Pg # 11, 12

167)

NCERT-XI, Pg # 18

The number of fungi from the provided list that have septate and branched mycelium with the absence of asexual spores is 1, specifically *Agaricus*.

168) NCERT XI, Pg. # 15

169)

NCERT-XI, Pg # 14, 15

170)

NCERT-XI, Pg # 14

171)

NCERT-XI, Pg # 90, 91

172) NCERT-XI Pg # 13

173)

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174) NCERT-XI, Pg. # 13

175)

NCERT-XI, Pg # 13,14,15

176) NCERT XI, Pg. # 14,15

177) NCERT-XI, Pg # 16

178) NCERT-XI, Pg # 17,18

179) NCERT (XI) Pg. # 17,18

180) NCERT-XI, Pg # 20,21